The Media Connectivity Association



OVERVIEW AND UPDATES · BROADCAST AND MULTICAST

Media delivery over 5G Broadcast

The 3GPP-based broadcast delivery system bridging OTT streaming and broadcasting

3GPP ETSI

OCTOBER 2026 – www.5g-mag.com

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About the project

The documents the 5G Broadcast project rests on

ETSI TS 103 720

Technical Specification

5G Broadcast System for linear TV and radio: the functions, reference points and receiver categories.

3GPP TR 36.976

Technical Report

The overall description of LTE-based 5G broadcast.

3GPP TS 26.346

Technical Specification

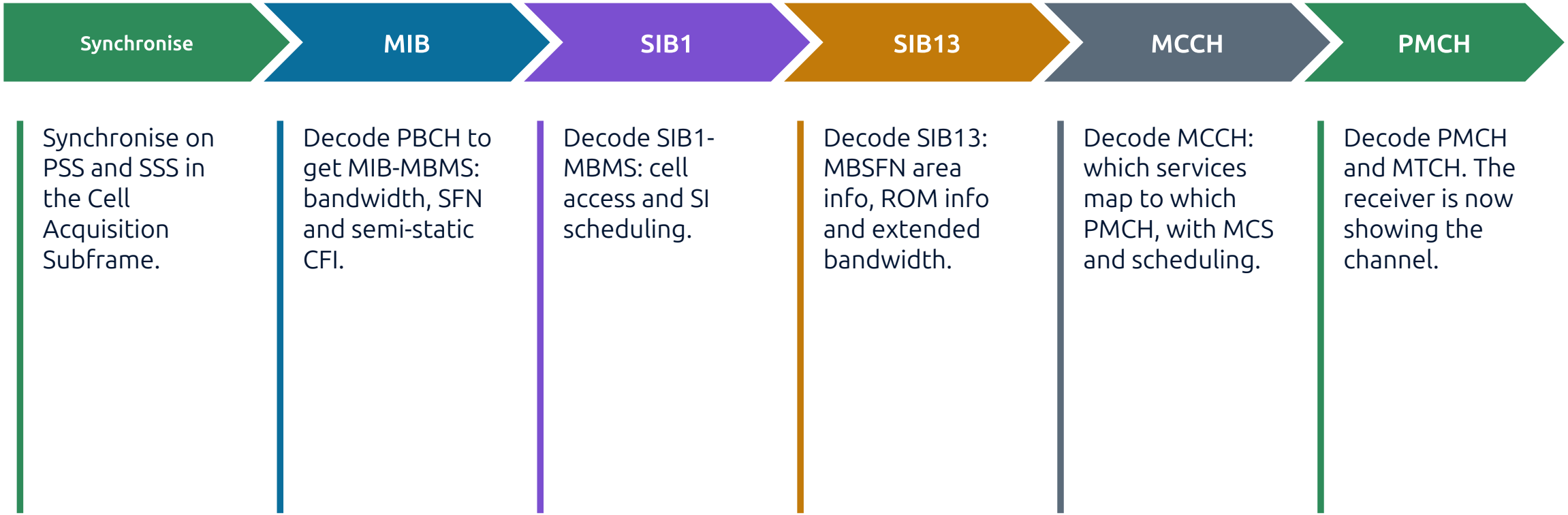
MBMS protocols and codecs. The MBMS application programming interface and URL are in 3GPP TS 26.347.

FIND OUT MORE **Technical Analysis** **Specifications** **Reference Tools**

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How a receiver tunes in

From nothing to a channel on screen, in the order the receiver acquires the signal



There is no SIM, no return channel and no prior connection to the network: the receiver bootstraps itself.

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Features by release

What each 3GPP release added to LTE-based 5G Broadcast; ETSI TS 103 720 V1.2.1 profiles Releases 14 to 17

Release 14 the enTV baseline

- MBMS-dedicated cell
- 1.25 kHz numerology
- Cell Acquisition Subframe
- 256-QAM on PMCH
- MIB-MBMS and SIB1-MBMS
- Shorter MCCH periods

Release 16 LTE-based 5G Terrestrial Broadcast

- 2.5 kHz and 0.37 kHz numerologies
- PDCCH Format 4
- PBCH repetition
- MBSFN area info
- Semi-static CFI
- Receive-Only Mode

Releases 17 and 18 bandwidths (17), UHF bands (18)

- 6, 7 and 8 MHz PMCH bandwidths (17)
- Band 107 (18)
- Band 108 (18)

Release 19 PMCH Phase 2, CAS muting, new bands; not in V1.2.1

- Time interleaving
- Frequency interleaving
- PMCH cyclic shift
- Extended MCH scheduling periods
- CAS muting
- Bands 112 and 113

The UHF bands are receive-only (downlink-only). Release 20 (work plan of 28 September 2026): Band 246 (satellite) bandwidths, unverified.

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How the specification grew

5G-MAG has maintained ETSI TS 103 720 since 2021, with changes agreed among members

December 2020

First publication, with Release 16 as baseline: Free-to-Air and Receive-Only Mode services, single frequency networks over large distances, mobility up to 250 km/h, DASH, CMAF and HLS, IP-based services and file delivery.

June 2023

Release 17 alignment: receiver requirements for network planning, 6, 7 and 8 MHz bandwidths and broadcast UHF spectrum, 5G Media Streaming over eMBMS, public warning and emergency alerts based on cell broadcast.

In progress

The Release 18 update, started in 2024: DVB-I over 5G Broadcast, band support including Band 108, 5G Broadcast as a public warning improvement, concurrent 5G Broadcast and unicast, 3GPP Service URLs, a RAN-only receiver profile and low-latency distribution.

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Receiver categories

Four categories in ETSI TS 103 720; the fourth, the Emergency Alerts Receiver, has no Feature Value

Base receiver

Feature value 27

The minimum profile, aimed at fixed indoor reception. The 1.25 kHz numerology is mandatory; other subcarrier spacings are optional.

Main receiver

Feature value 28

Supports all five numerologies: 15, 7.5, 2.5, 1.25 and 0.37 kHz. The target for mobile and portable devices.

5GMS receiver

Feature value 30

Meets all the receiver requirements plus the 5G Media Streaming requirements, for 5GMS over LTE-based 5G Broadcast.

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The receiver chain

From the radio signal to a media player, as the Reference Tools receive it



The 5G Broadcast signal arrives as live I/Q data from an SDR, or from a prerecorded sample file.

Converts the signal into multicast IP packets on its output.

Decodes the FLUTE-delivered content from the multicast traffic and serves it to a media player.

Plays the stream in the browser with dash.js or hls.js.

IMPLEMENT

Repositories: transmitter and receiver

Transmitter, modem, middleware and web interface, from the descriptions of the repositories

REPOSITORY	WHAT IT IS	COVERS	LATEST RELEASE	
rt-mbms-tx	a standalone 5G Broadcast transmitter	Transmitter	● v1.0.0 21 Jun 2023	GitHub
rt-mbms-tx-for-qrd-and-crd	an MBMS-enabled eNodeB extended as a transmitter for QRD and CRD devices	Transmitter, QRD and CRD devices	● ea-1.0.0 5 Sep 2024	GitHub
rt-mbms-modem	converts an SDR or sample-file signal into multicast IP packets	Modem, SDR input, Sample files	● v1.2.1 30 Jun 2022	GitHub
rt-mbms-mw	decodes FLUTE content from the Modem and serves it from a web cache	Middleware, Web cache	● v0.10.0 8 Dec 2022	GitHub
rt-mbms-mw-android	an implementation of the MBMS Middleware for Android	Middleware, Android	● 1.0.0 5 Sep 2024	GitHub
rt-wui	a web interface that controls the Modem and Middleware and plays HLS	Web interface, HLS playback	● v0.1.0 8 Dec 2022	GitHub

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● released ○ no release yet

More information at 5g-mag.com/tech/5g-broadcast

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Repositories: core, applications and tools

The BM-SC, the Gateway, the Client, the applications and the tools, from the repository descriptions

REPOSITORY	WHAT IT IS	COVERS	LATEST RELEASE	
rt-mbms-bmsc	the BM-SC: ingests content, announces it and delivers it with FLUTE	BM-SC	none yet	GitHub
rt-mbms-gw	takes content from the BM-SC and delivers it onto the broadcast bearer	MBMS Gateway	none yet	GitHub
rt-mbms-client	reads the announcement, reassembles FLUTE objects, republishes HLS or DASH	MBMS Client	none yet	GitHub
rt-mbms-application	a web app that plays the MBMS Client output and shows radio state	MBMS-Aware Application	none yet	GitHub
rt-mbms-application-provider	a provider portal that provisions services and sessions on the BM-SC	Application Provider	none yet	GitHub
rt-libflute	FLUTE version 1, which carries files over the broadcast bearer	FLUTE	0.12.4 1 Sep 2026	GitHub
rt-mbms-examples	examples that use other 5G-MAG repositories, such as FLUTE ffmpeg	Examples, FLUTE ffmpeg	1.1.0 5 Nov 2024	GitHub

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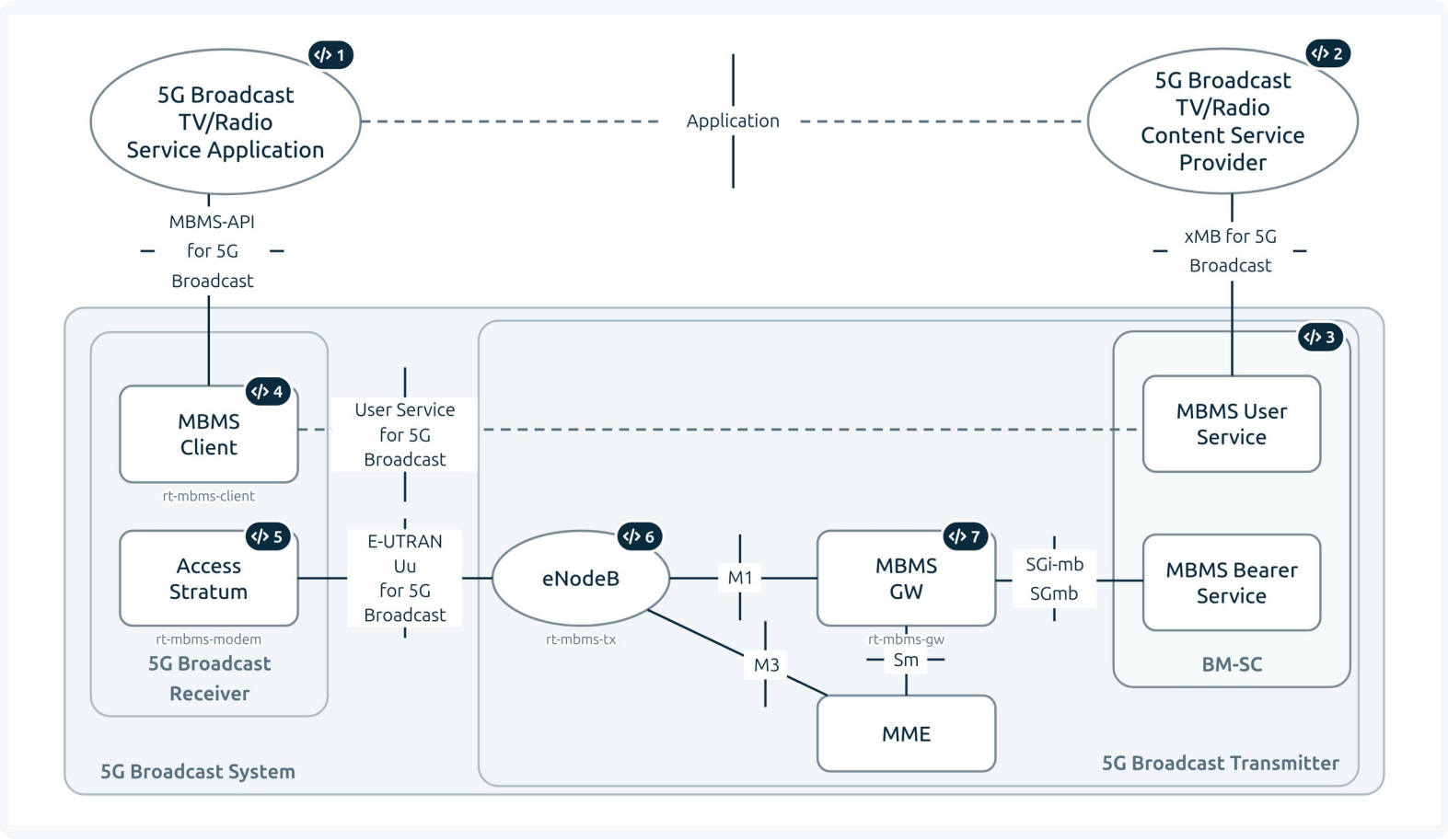
● released ○ no release yet

More information at 5g-mag.com/tech/5g-broadcast

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Repositories in the architecture

The Reference Tools on the specification figure, as the figure generator draws them



PROVIDED BY THE REFERENCE TOOLS

1. **rt-mbms-application**
2. **rt-mbms-application-provider**
3. **rt-mbms-bmsc**
4. **rt-mbms-client**
   used by 1
5. **rt-mbms-modem**
   used by 1
6. **rt-mbms-tx**
   used by 3, 7
7. **rt-mbms-gw**
8. **rt-libflute**
   dependency, used by 3, 4
9. **rt-common-shared**
   auxiliary library, used by 4

Boxes without a badge are not mapped to a repository. The MME has no repository.

See it and try it

A test setup with an SDR and phones, and the middleware app playing a live stream



SDR and phones, test setup



Middleware app

From the Reference Tools tutorials: seamless switching between broadcast and unicast on Android, and SDR playback.

Latest updates and roadmap

Status on 22 October 2026

RELEASED

rt-libflute 0.12.4

1 Sep 2026. Option to finish sending all files when the transmitter is deactivated; fixes unbounded FDT growth.

rt-mbms-examples 1.1.0

5 Nov 2024. Adds configuration changes and sample scripts for the IBC 24 demo.

rt-mbms-mw-android 1.0.0

5 Sep 2024. Seamless switching between 5G broadcast and unicast delivery on Android, with HLS support.

IN DEVELOPMENT

ETSI TS 103 720 update

The update of the specification for 3GPP Release 18 is in progress.

BM-SC and application provider

xMB interface fields reworked to the specification, real Raptor FEC and a stricter FLUTE profile in the BM-SC.

MBMS Client announcement

The MBMS Client gains every client-side way of acquiring the service announcement, and a broadcast demo chain.

PLANNED

Release 17 to 19 transmitter

Wider bandwidths, new UHF bands, CAS muting and interleaving in the QRD and CRD transmitter.

Android playback modes

DASH and RTP playback, and seamless switching with DASH, in the MBMS Android Middleware.

Docker setup

Container images for the transmitter and the client-side components.

MERGED SINCE V1.0.0 61 pull requests in 8 repositories

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